Voi / Portfolio Lab 7

Frozen paper evidence pack · 28 October 2026

Research question: under what customer, operational and competitive conditions do the two offers deserve investment—and what evidence changes that decision?

Illustrative starting cohort: 10,000 active customers. Horizon: 36 months. Annual discount: 10%. No terminal value. Configuration 4fc8f515adf767ee.

Decision under the frozen assumptions

Investment	Δ contribution	Launch	Peak cash	Payback
No investment	0 kr	0 kr	0 kr	None
Hub & Book	−1.30 MNOK	409,733 kr	1,435,637 kr	None
Voi Together	−0.62 MNOK	292,074 kr	668,770 kr	None
Both offers	−1.78 MNOK	669,104 kr	1,953,210 kr	None

No investment under these conditions. Current costs and demand assumptions do not support launch at the selected scale. This is a conditional finding; it does not reject the underlying customer needs. Fixed costs are retained and have not been tuned to produce a positive answer.

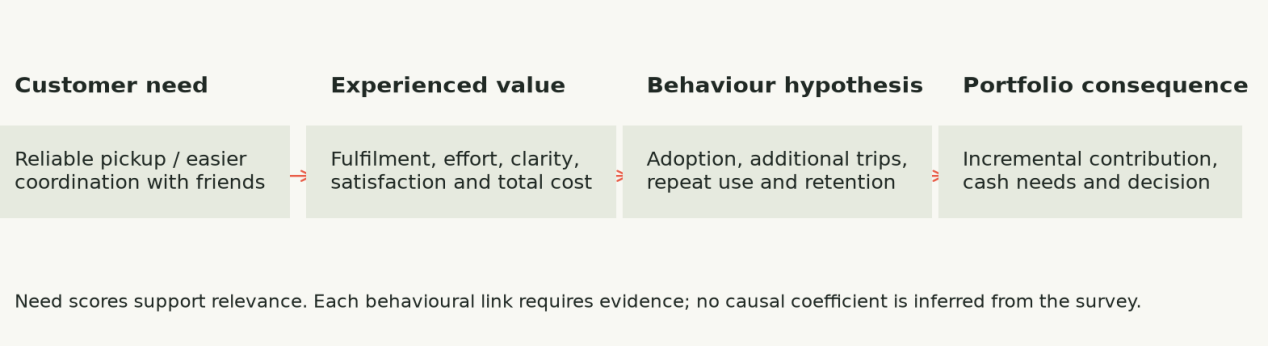
The 750,000 kr launch budget is illustrative, not specified by Assignment 2. The total funding limit is unspecified. Launch affordability does not confirm that later cash needs can be financed.

Uncertainty remains conditional

Investment	Simulated mean	5th–95th percentile	Loss share
No investment	0 kr	0 kr to 0 kr	0.0%
Hub & Book	−1.33 MNOK	−1.45 MNOK to −1.22 MNOK	100.0%
Voi Together	−0.62 MNOK	−0.62 MNOK to −0.61 MNOK	100.0%
Both offers	−1.81 MNOK	−1.93 MNOK to −1.70 MNOK	100.0%

3,000 paired draws; seed 2026. Triangular ranges and dependencies are assumptions. Loss shares are shares of model draws, not observed probabilities of business success. No calibrated value of pilot learning is included.

Customer value and the calculation



CS need scores: battery 6.68/7; nearby supply 6.23/7. SLS: travelling with friends 6.10/7; evening use 6.71/7. The 72-person convenience sample supports descriptive segment interpretation, not population weights or response rates.

Δ contribution = discounted monthly offer cash minus matching no-offer cash, less setup. The two runs face the same external demand path. Guests enter later customer states; no up-front CLV is added. Usage states are occasional, frequent, subscriber and inactive, not relationship-quality categories.

Blended net revenue benchmark: EUR178.2m /115.815m rides ×11.7173 = 18.0290 NOK/ride. Group average proxy; PAYG revenue is reconciled after net Pass fees. Historical 399 kr Pass is an assumption. Ordinary ride cost is an average proxy, not measured marginal Oslo cost.

Retention is a bounded relative reduction among exposed customers before exposure weighting. Retention, repeat, upgrade and spillover start at zero. Booking-driven spillover requires actual placements. Adoption, additionality, acquisition and operating assumptions still need measurement.

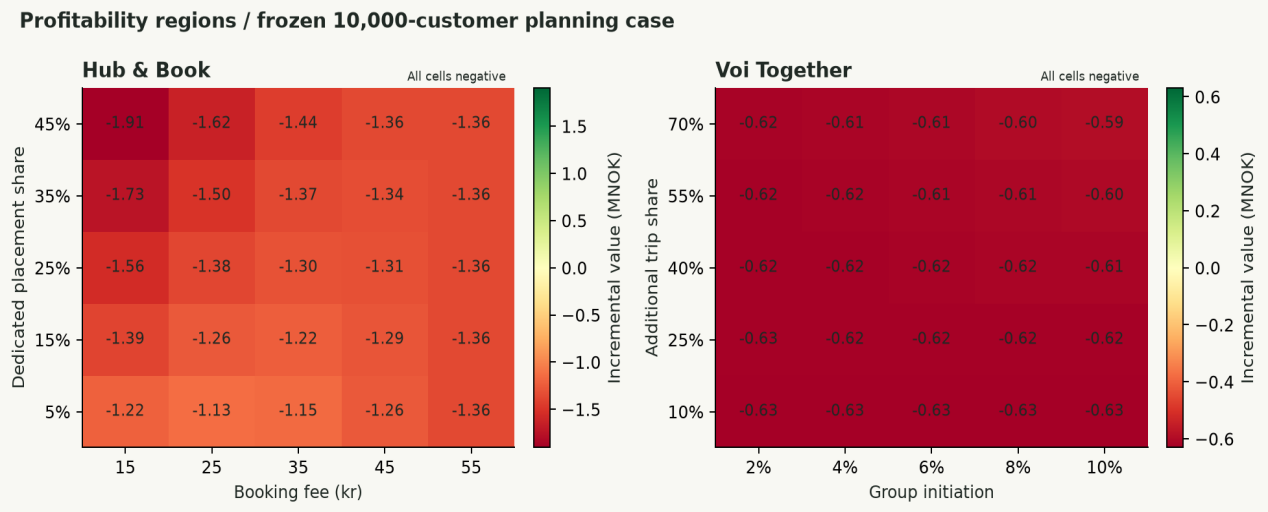
Default price-response coefficients (booking 1, group incentive 2) are planning hypotheses. Zero and alternative responses are compared explicitly. No survey question estimates these coefficients or purchase of the exact proposed service.

Competitive protection is a hypothesis

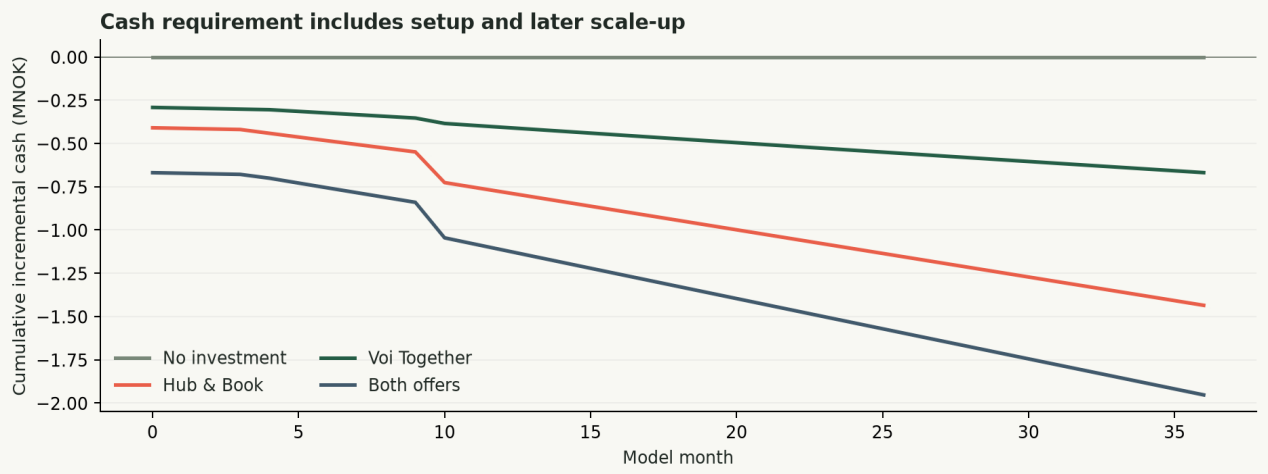
Johnson & Selnes (2004) and Selnes & Lanseng (2024) support the dynamic portfolio logic. Burnham, Frels & Mahajan (2003) distinguishes procedural, financial and relational switching costs. Satisfaction, retention and switching barriers must be measured separately. Rival copying, relative pricing and fading benefits change existing mechanisms; no second defence premium is added.

Mobility literature informs mechanisms and cost categories. Kabra, Belavina & Girotra (2020), doi:10.1287/mnsc.2019.3407, concerns bike-share availability; its coefficient is not an estimated Voi treatment effect. Complete dated references and input statuses accompany this PDF in the source register.

Where the answer changes



Numbers are MNOK for each offer alone. At this cohort size there is no profitable region within the tested grid. Do not extend prices or adoption simply to manufacture a crossing. The interactive maps also identify the best feasible portfolio at each cell.



Cash is undiscounted. Setup is paid at month 0, with later scale-up and operating costs. A positive discounted result would not by itself establish sufficient cash funding. The JSON and monthly CSV reconcile these figures exactly.

What the archived result means

The archived v6.6 +4.18 MNOK case uses an assumed Oslo-wide scale, nonzero relationship effects and old mechanics. It remains reproducible but is not directly comparable with the smaller v7 pilot planning case. The v7 hypothesis comparison holds current scope and delivery assumptions constant.

Pilot, delivery and control

Offer	Launch	Ramp complete	Retention window	Gate
booking	4	7	8–13	9
group	5	7	8–13	9

Six complete post-ramp retention months extend to month 13; the month-9 gate cannot validate retention. Operational and retention decisions must be distinguished. Direct rollout has no experimental gate.

Primary economic outcome: contribution per assigned eligible customer against a concurrent control, including non-users. Stable hub-zone clusters limit supply spillover; invitation networks stay together. Record contamination and deduplicate guests. Per-booker comparisons require additional identification assumptions.

Price-test conversion, placements and crew minutes, fulfilment, battery, displaced rides, additional group trips, guest activation, customer effort, satisfaction, competing-app use and switching reasons. Agree guardrails before launch. Exclude sunk setup when deciding continuation.

Contribution variance, cluster counts, ICC, attrition and minimum worthwhile effect are missing. Therefore no study precision is asserted. The calculator provides an approximate normal-based planning result only after these inputs are supplied; few clusters need specialist validation.

Owner	Accountability
Country manager / pilot owner	Sponsor, cross-functional decisions, guardrails and review.
Product / Operations	Truthful booking terms, app flow, capacity, fulfilment and crew records.
Commercial / Analytics	Price tests, stable assignment, incremental outcomes and competition evidence.
Finance	Incremental cost boundaries, budget, cash needs and continuation economics.

Forecasting decision

No compatible historical scooter trip series was obtained in this audit. This is a data-availability finding, not proof that none exists. Explicit seasonality scenarios are used.

Entur GBFS describes availability/status; Oslo City Bike has bike trip history; Ruter supplies annual scooter totals. No comparable scooter time series was obtained. Prophet stays off. Qualified completed-trip data must beat a seasonal baseline at three historical origins before adoption. Imported demand indices affect both sides once; longer projections remain scenarios.

Use in the term paper

Required section	Use and interpretation
Executive Summary	Conditional recommendation, principal uncertainty and next decision.
Situation Analysis	Survey needs, current capabilities and competitive copyability; no population inference.
Financial Goals	Incremental discounted contribution, null/break-even conditions, horizon and payback.
Strategic Initiatives	Compare offer variants; explain customer value and the implementation mechanism.
Budget	Launch investment, recurring work, scale-up and peak cash; budget is illustrative.
Implementation & Control	Accountable owners, assigned-customer outcomes, measurement windows and stop rules.

The maximum ten-page core report must stand alone. Explain the recommendation, conditions, customer value and limitations in the paper. Interactive submission acceptance is not assumed. The earlier organizational-change draft still discusses Ruter; reconcile the consolidated paper before submission.

Reproducibility and limits

The pack contains full configuration/results JSON, monthly CSV, sources-and-assumptions CSV, methods, pilot protocol, report mapping and three reusable SVG/PNG figures. The configuration hash identifies the frozen case. UI edits do not silently alter these files.

Three real first-time readers have not yet completed the five-minute usability test. Automated browser checks cannot substitute for this. Use the included protocol before final release acceptance.

No validated causal treatment effects, representative segment weights, observed Oslo unit economics, fitted customer forecasts or calibrated experiment value are claimed. Preserve negative findings and collect decision-changing evidence.

Selected references

Voi Technology AB (2026). Annual Report and Sustainability Statement 2025, Note 2.

Our survey dataset VOI_Dataset_Original.sav (n=72, convenience sample), four-cluster solution QCL_4.

Johnson, M. D. & Selnes, F. (2004). Customer portfolio management. Journal of Marketing 68(2), 1–17.

Burnham, T. A., Frels, J. K. & Mahajan, V. (2003). Consumer Switching Costs: A Typology, Antecedents, and Consequences. Journal of the Academy of Marketing Science 31(2), 109–126. Publisher abstract checked; no Voi coefficient transferred.

Customer Centricity, Lecture 5: Customer satisfaction, switching costs and customer loyalty, slides 12–14, 18–25. Supplied course material; conceptual guidance, not Voi effect estimates.

Kabra, Belavina & Girotra (2020). Bike-share systems: Accessibility and availability. Management Science 66(9), 3803–3824. doi:10.1287/mnsc.2019.3407.

Selnes & Lanseng (2024). Customer Portfolio Management. In Marketing Management: A Customer-Centric Approach, pp. 141–186. Supplied course chapter.